

Solution Code - PL SQL programming - Functions

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: solution.sql

```
1 | -- Scenario 1: CalculateAge
2 | CREATE OR REPLACE FUNCTION CalculateAge(p_dob DATE) RETURN NUMBER IS
3 | BEGIN
4 |     RETURN TRUNC(MONTHS_BETWEEN(SYSDATE, p_dob) / 12);
5 | END;
6 | /
7 |
8 | -- Scenario 2: CalculateMonthlyInstallment
9 | CREATE OR REPLACE FUNCTION CalculateMonthlyInstallment(
10 |     p_amount NUMBER,
11 |     p_rate NUMBER,
12 |     p_duration_years NUMBER
13 | ) RETURN NUMBER IS
14 |     v_monthly_rate NUMBER;
15 |     v_months NUMBER;
16 | BEGIN
17 |     v_monthly_rate := (p_rate / 100) / 12;
18 |     v_months := p_duration_years * 12;
19 |     IF v_monthly_rate = 0 THEN
20 |         RETURN p_amount / v_months;
21 |     ELSE
22 |         RETURN p_amount * (v_monthly_rate * POWER(1 + v_monthly_rate, v_months)) / (POWER(1 +
v_monthly_rate, v_months) - 1);
23 |     END IF;
24 | END;
25 | /
26 |
27 | -- Scenario 3: HasSufficientBalance
28 | CREATE OR REPLACE FUNCTION HasSufficientBalance(
29 |     p_acc NUMBER,
30 |     p_amount NUMBER
31 | ) RETURN BOOLEAN IS
32 |     v_bal NUMBER;
33 | BEGIN
34 |     SELECT Balance INTO v_bal FROM Accounts WHERE AccountID = p_acc;
35 |     RETURN v_bal >= p_amount;
36 | EXCEPTION
37 |     WHEN NO_DATA_FOUND THEN
38 |         RETURN FALSE;
39 | END;
40 | /
```